# PHAEMA test programme

Nov 10, 2026 · @Stuart

## How to use this programme

This programme tests every section of PHAEMA in about 12 working days, using the demonstration data in two fictional districts. Each test below is a set of instructions for a person to follow in PHAEMA: an ID, steps, the expected result and a Result column to fill in. Tests marked (IT) need someone who can type commands on the computer that runs PHAEMA.

**Where to test.** Either use the public test site at demo.phaema.co.uk, or install a private copy for your council with the installation guide in the code repository (`deploy/README.md`). On the public site, ask for sign-ins with **Request a test account** (the PHAEMA project sets them up), and note that all data is reset every 48 hours: run each section within a day, record results as you go, and do not rely on records surviving overnight. Tests that need an Administrator account, or that change other people's work (sections U, Y and Z), are best run on a private copy.

**Test data.** Load it with `python manage.py load_demo_data --reset` (an IT task). It holds 72 properties, 47 agreements, 56 cases, 22 repair jobs, 59 certificate requirements, 35 valuations and two service charge schedules. All names, addresses, numbers and documents are invented.

| District | References | Good for testing |
|---|---|---|
| Exampleton | DM-E01, DM-S01 to DM-S11 | Riverside Industrial Estate, Market Square shops, Town Hall, Enterprise Centre (leased in, sub-let), a confidential property, surplus school site |
| Northfield | DM-N01 to DM-N12 | Business park, The Precinct shops, library and leisure centre (compliance), former bus depot (hazards, surplus), telecoms site, market hall (turnover rent), farm, Innovation Hub (leased in) |

Agreements are DL-0001 to DL-0200 (Exampleton) and DL-0501 to DL-0531 (Northfield). Dates are set relative to the day you load the data, so reviews, breaks and certificates always fall due at a realistic mix of times. Reload before each test cycle so everyone starts from the same position.

**Projects.** Three fictional projects: PRJ-2026-001 (a development at Northfield Business Park), PRJ-2026-002 (a confidential depot site acquisition) and PRJ-2026-003 (a disposal of two surplus sites). Section PJ uses them.

**Demonstration people.** Surveyors Alex Morgan, Priya Shah, Owen Jones, Lena Chen and Rob Baker are supervised by Sam Hughes (Head of Estates); Jo Patel is a legal contributor. They cannot sign in. Use them to test assignment, approvals and My work, viewed as an administrator.

**Test accounts.** In Manage data, create one account per role for each tester, with a password: an Administrator, two Surveyors (one supervising the other), a Contributor and a Viewer. Give one Surveyor some properties as managing surveyor. Put every account in all five access areas to start with; section Z narrows them. Give every account its own email address. Administrators are asked to set up two-step sign-in the first time they sign in (test A9), so have an authenticator app ready on a phone. Never use real people's details.

**Before you start:** switch on every module in Manage data, Module settings (rent ledger, service charges, maintenance, compliance, asset valuation). Some tests switch one off and back on.

## Programme

Run the phases in order: later sections depend on the register, leases and cases working. Two testers can split each phase by section.

| Days | Phase | Tests |
|---|---|---|
| 1 | Set-up and smoke test: load data, create accounts, sign in as each role, check every menu item opens | A, B |
| 2–3 | The register: properties, map, titles, parties, agreements, lease events and reminders | C, D, E, F |
| 4–5 | Casework: cases, approvals, heads of terms and 1954 Act, standards checks, document library, projects | G, H, I, J, PJ |
| 6–7 | Income: rent ledger and service charges | K, L |
| 8–9 | Buildings: maintenance, compliance and Golden Thread, asset valuation | M, N, O |
| 10 | Data in and out: reports, imports and exports, open data, API, change log, settings | P, Q, R, S |

| Days | Phase | Tests |
|---|---|---|
| 11 | Cross-cutting: accessibility, security and permissions, browsers and devices, performance, access areas and cover | T, U, V, W, Z |
| 12 | Hand over work (changes demo data, so run last and reset), retest fixed defects, run the automated suite, sign off | Y, X |

**Start when:** the demo data loads without errors, every tester has their accounts, and the automated test suite passes (`python3 -m pytest`).

**Finish when:** every test has a result; no Critical or High defects are open; Medium defects have an agreed fix date or are accepted; the automated suite passes on the final build.

## Recording results

In each test's Result column write **P** (pass), **F** (fail, with a defect number), **B** (blocked by another defect) or **N/A** (module not in use), plus your initials and the date.

| Severity | Meaning | Example |
|---|---|---|
| Critical | Data lost or wrong, a security hole, or a whole section unusable | A Viewer can edit a lease; a rent charge is posted twice |
| High | A main task cannot be done, with no workaround | Cannot record a rent review outcome |
| Medium | A task works but wrongly or awkwardly, with a workaround | A report total is missing; a filter ignores one option |
| Low | Wording, layout or cosmetic | A typo; misaligned column |

For each defect record the test ID, the page address, what you did, what you expected, what happened, a screenshot, and your browser. Raise it as an issue on GitHub (phaema/phaema-app) with the label `testing`, or in the defect log at the end of this document.

# Tests A–D: access, dashboard, register and map

## A. Sign-in and roles

| ID | Test | Steps | Expected | Result |
|---|---|---|---|---|
| A1 | Sign in and out | Sign in with each test account; select Sign out | Dashboard opens; the header shows your name and role; sign out returns to the sign-in page | |
| A2 | Wrong password | Sign in with a wrong password | Clear error; no hint whether the username exists | |
| A3 | Viewer is read-only | As Viewer, open a property, an agreement and a case | No edit buttons or forms; posting a change by URL is refused | |
| A4 | Contributor | As Contributor, open a case | Can add notes, emails and documents; cannot move stages or decide | |
| A5 | Surveyor | As Surveyor, change a property you manage, then one you do not | First saves; second is refused | |
| A6 | Administrator | As Administrator, open Import, Manage data and Hand over work | All three open; other roles do not see these links | |
| A7 | Pages need sign-in | Signed out, open a property address and an uploaded file address | Sent to sign-in; files are never served | |
| A8 | Sign in with email | Sign in typing your email address in place of the username, then again in capital letters | Signs in both times; a wrong password gives the same error as A2 | |

| ID | Test | Steps | Expected | Result |
|---|---|---|---|---|
| A9 | Two-step set-up | Sign in with the Administrator test account for the first time; set up two-step sign-in with an authenticator app; note the recovery codes; sign out and in again | Asked to set it up before anything else; the next sign-in asks for a 6-digit code after the password | |
| A10 | Two-step codes | Enter a wrong code five times; sign in again using a recovery code; try the same recovery code once more | Five wrong codes end the attempt; a recovery code works once only | |
| A11 | Forgotten password | On the sign-in page select Forgotten your password?; enter your email address; open the emailed link and choose a new password; open the link again. Needs email set up on the server | The email names your username; the new password is saved; you are not signed in until you sign in yourself, with your code if you use two-step sign-in; the second use says the link has expired | |
| A12 | Forgotten password, no account | Enter an address that has no sign-in, then the username of an account that is not Active | The same Check your email page each time; no email is sent | |
| A13 | Change password | Select Change password in the menu: enter a wrong current password, then the right one with a new password of 12 or more characters | First refused; second saved; the new password signs you in | |

## B. Dashboard, My work and page settings

| ID | Test | Steps | Expected | Result |
|---|---|---|---|---|
| B1 | Landing tab | Sign in as Surveyor, then as Viewer | Surveyor lands on My work; Viewer on Whole estate | |
| B2 | Remember tab | As Surveyor choose Whole estate, sign out and in | Whole estate opens | |
| B3 | Whole estate figures | Check property count, vacant units, rent roll and lease event counts against the Properties and Events lists | Figures agree; each number links to the matching filtered list | |
| B4 | My work content | As Administrator, Show the work of Lena Chen | Needs attention, Coming up (90 days), open cases, properties (Northfield Business Park and others), repairs, compliance and arrears panels | |
| B5 | Viewing others | As a Surveyor, add ?view=mine&person= with another surveyor's number | Refused unless you supervise them | |
| B6 | Hide and show panels | Hide two panels, reload; use Hide all and Show all | Hidden panels stay hidden after reload; buttons toggle all | |
| B7 | Styles | Switch Style to Clear, then Compact, then Classic | Page changes look; choice kept across pages | |
| B8 | Themes and text size | Try Dark, High contrast and Largest text with the Clear style | All text readable; nothing overlaps or is cut off | |
| B9 | Panels without JavaScript | Turn JavaScript off and open the dashboard | Every panel is open and readable | |

| ID | Test | Steps | Expected | Result |
|---|---|---|---|---|
| B10 | Tenancies at will | Whole estate: select the highlighted tenancies at will in place figure | Opens the agreements list showing only live tenancies at will (DL-0040 is one); the number listed matches the figure | |
| B11 | Expiries by 1954 Act status | Under Lease events, compare the lease expiries row with the Contracted in and Contracted out rows beneath it; select each number | The rows beneath add up to the lease expiries row; each number opens the matching events | |
| B12 | Time is of the essence | Under Lease events, select the links in the red Time is of the essence notice | Lists the open rent reviews where time is of the essence, including DL-0001; the counts match the notice | |
| B13 | Documents to review | Under Lease events, select the numbers on the Lease documents to review row | Opens Documents to review with the overdue or coming review dates; each document links to its agreement | |

## C. Properties

| ID | Test | Steps | Expected | Result |
|---|---|---|---|---|
| C1 | List and filters | Properties: filter by type Unit, occupation Vacant, surveyor, and search "Precinct" | Results match every filter; counts shown | |
| C2 | Hierarchy | Open DM-N01 Northfield Business Park | Shows its two blocks and ten units; breadcrumbs work both ways | |
| C3 | Floor areas | Open DM-E01-B01 | Area breakdown adds up the units; measurement basis shown | |

| ID | Test | Steps | Expected | Result |
|---|---|---|---|---|
| C4 | Photos | Open DM-N01-U04 and click each thumbnail | Main photo and caption change; works by keyboard | |
| C5 | Site hazards | Open DM-N06 Former bus depot | Hazard warning shown at the top | |
| C6 | Add and edit | As Administrator add a property under DM-N02, then edit it | Saved; appears in the list and in the change log | |
| C7 | Confidential site | Find DM-S09 | Marked excluded from open data | |
| C8 | Portfolios | As Administrator, Manage data, Portfolios: add one; choose it on a property in Edit this property; filter the register by it; try to delete it | The property shows the portfolio; the filter and the Portfolio column work; a portfolio with properties in it cannot be deleted | |
| C9 | Estate strategy and purpose review | Edit a property: enter an Estate strategy section and a Holding purpose review date in the past; filter the register to purpose reviews that are overdue | Both are shown on the property; the property appears under the overdue filter | |

## D. Map and titles

| ID | Test | Steps | Expected | Result |
|---|---|---|---|---|
| D1 | Map loads | Open Map; zoom to Exampleton, then Northfield | Boundaries shown for both districts; legend matches colours | |
| D2 | Select on map | Click a boundary | Property name and link shown | |
| D3 | Draw from parcels | On DM-S06 join the two registered parcels into one boundary | Boundary saved; site area recalculated | |

| ID | Test | Steps | Expected | Result |
|---|---|---|---|---|
| D4 | Titles | Open title ZZ100014 (DM-N06) and the unregistered title for DM-N05 | Title class (qualified, unregistered), deeds location and linked properties shown | |
| D5 | Title documents | Open the official copy on ZZ100001 | PDF opens; only signed-in users can open it | |

# Tests E–J: agreements and casework

## E. Parties and agreements

| ID | Test | Steps | Expected | Result |
|---|---|---|---|---|
| E1 | Parties | Open Northfield Group plc | Shows it guarantees DL-0503; contact details and finance reference shown | |
| E2 | Agreement types | Open DL-0505 (lease), DL-0523 (licence), DL-0525 (farm business tenancy), DL-0522 (telecoms) | Each shows the right type, term, rent and frequency | |
| E3 | Leased in | Open DL-0530 and DL-0100 | Shown as held by the council; rent counted as paid, not received | |
| E4 | Rent schedule | Open DL-0034 | First-year concession then full rent; current rent correct for today | |
| E5 | Breaks | Open DL-0502 and DL-0519 | Break date, who can break, notice period and last notice date shown | |
| E6 | Holding over | Open DL-0507 | Status Holding over; appears in the holding-over report | |
| E7 | New agreement | As Surveyor create a lease of DM-N01-U04 with a 5-year review and a break | Saved; review, break and expiry events created and assigned to the managing surveyor | |

| ID | Test | Steps | Expected | Result |
|---|---|---|---|---|
| E8 | Deposit | Open DL-0502 | Deposit amount, where it is held and who gets the interest shown | |
| E9 | Agreement flags | Agreements: filter by each flag in turn (time is of the essence, rent demands on hold, arrears passed to debt collectors or legal, has its own billing contact) | DL-0001 (essence, billing contact), DL-0031 (on hold) and DL-0024 (arrears passed on) appear under the right flags, and each is tagged on its own page | |
| E10 | Billing contact | Open DL-0001; then on an agreement with no billing contact select Edit this agreement and add only a billing email | DL-0001 shows its own billing contact; the other shows the tenant's main contact with your email in place of theirs | |
| E11 | Document review date | On an agreement add a document with a date for review in the past | Listed on the agreement with its review date marked overdue; appears in Documents to review (B13) | |
| E12 | Tenant's insurance certificate | Open DL-0021; then an agreement where the tenant insures and no certificate is held, and select Start a case to get the tenant's insurance certificate | DL-0021 shows the certificate held; the other shows a warning; the new case is linked to the agreement and has its tasks | |

## F. Lease events and reminders

| ID | Test | Steps | Expected | Result |
|---|---|---|---|---|
| F1 | Events list | Events: filter by type, overdue, next 6 months, assigned to me | Correct events; overdue in red | |

| ID | Test | Steps | Expected | Result |
|---|---|---|---|---|
| F2 | Record a review | Settle an open rent review at a new rent | Event done; rent schedule gains the new rent from the review date | |
| F3 | Lapsed breaks | Find a break date in the past | Shown as lapsed, never overdue | |
| F4 | Reminders (IT) | Run `python manage.py send_reminders` | Emails list each surveyor's due events at the configured lead times | |
| F5 | Calendar feed | Subscribe to your calendar link in a calendar app | Your events appear with correct dates | |
| F6 | Estimated rental value | Open the rent review on DL-0001 and the lease expiry on DL-0004; change the estimated rental value on one; then open a break event | The estimated rental value is shown against the current rent and your change is saved; a break event has no estimated rental value box | |
| F7 | Time of the essence on events | Edit an agreement and tick Time is of the essence in the rent review section; open Events filtered to rent reviews where time is of the essence; open My work for its surveyor | Its open rent review is listed and tagged, and marked on My work; the dashboard count (B12) rises by one | |

## G. Cases and approvals

| ID | Test | Steps | Expected | Result |
|---|---|---|---|---|
| G1 | Case list | Cases: filter by stage, owner, past target, waiting for approval | Correct cases; RAG tags match target dates | |
| G2 | New consent case | Open a consent to assign on DL-0511; acknowledge; mark the application complete | Target date set from the application date (28 days); tasks created | |

| ID | Test | Steps | Expected | Result |
|---|---|---|---|---|
| G3 | Tasks | Tick tasks done; add your own task with a due date | Saved with who and when; overdue tasks show on My work | |
| G4 | Correspondence | Log a letter; file an email (.eml) with an attachment | Both in the timeline; attachment stored on the case | |
| G5 | Confidential notes | Add a confidential note; view as Contributor | Contributor sees a count of hidden notes, not the text | |
| G6 | Approval gate | Try to move a consent case to Decision | Replaced by Ask for approval | |
| G7 | Ask, return and approve | Ask for approval; as supervisor Return without comments, then with; ask again; Approve | Returning needs comments; emails sent each way; history kept on the case | |
| G8 | No self-approval | As the requester, post an approval | Refused | |
| G9 | Reassign | As Administrator send a waiting approval to someone else | New approver sees it on My work | |
| G10 | Decide and close | Record Granted with conditions, then close | Conditions need written reasons; case closes | |
| G11 | Fee | Raise a case fee | Charge created on the rent ledger, linked to the case | |
| G12 | Event cases | Open the case for an upcoming rent review | Linked to the lease event; closes when the event outcome is recorded | |

## H. Heads of terms and the 1954 Act

| ID | Test | Steps | Expected | Result |
|---|---|---|---|---|
| H1 | Heads of terms | Open the Studio 2 new letting case (Northfield); fill all 16 items | Counter rises to 16 of 16; current lease terms shown as hints for renewals | |
| H2 | Unrepresented tenant | Mark the tenant unrepresented without a Code date | Decision blocked until the RICS Code date is given | |
| H3 | Section 25 notice | On the Unit 7 renewal case check the notice dates | Court deadline and counter-notice dates calculated; ground (f) opposition shown | |
| H4 | Section 26 request | On Brookside Joinery's renewal case change the date served | Deadlines recalculate; warnings for impossible dates | |

## I. Standards checks on cases

| ID | Test | Steps | Expected | Result |
|---|---|---|---|---|
| I1 | Conflict of interest | Try to ask for approval before confirming conflicts; then declare one with details | Blocked first; declaration saved and shown to the approver | |
| I2 | Dilapidations | Open the Unit 4 dilapidations case; enter a settlement | Schedule £68,500, response £31,000; gap to the schedule shown | |
| I3 | MEES warning | Open a new letting case for DM-N01-U04 (EPC G) and DM-N12-U02 (no EPC) | Warning about the rating, and about the missing EPC | |
| I4 | Before you visit | Open a case, job and valuation on DM-N04 Leisure Centre | Panel lists the pool plant room hazard, asbestos information and open actions | |

### J. Document library

| ID | Test | Steps | Expected | Result |
|---|---|---|---|---|
| J1 | Browse | Documents: browse by category | All templates listed with their legal source | |
| J2 | Fill from a case | From a consent case create a consent approval letter | Tenant, premises, lease date and case reference filled in; gaps highlighted | |
| J3 | Export to Word | Download the letter as Word | Opens in Word with letterhead, page numbers; unfilled items highlighted yellow | |
| J4 | Prescribed forms | Create a section 25 notice | Particulars only, with a warning to use the prescribed form | |
| J5 | Add a template | As Administrator add a new template with merge fields | Available to surveyors; fields fill correctly | |

## Tests K–O: income and buildings

### K. Rent ledger

| ID | Test | Steps | Expected | Result |
|---|---|---|---|---|
| K1 | Rent account | Open DL-0506 Northfield Joinery Workshop | Monthly charges, receipts and running balance; last two months unpaid | |
| K2 | Arrears | Rents, Arrears | DL-0506, DL-0512, DL-0518 and Exampleton's DL-0016 and DL-0024 listed, aged into 1–30, 31–60, 61–90 and 90+ days | |
| K3 | Part payment | Open DL-0514 Northfield Bakery | Latest charge part paid (about 40%) | |
| K4 | Demand run | Issue the next quarter's demands; issue it again | Charges created once only, with VAT where chargeable | |

| ID | Test | Steps | Expected | Result |
|---|---|---|---|---|
| K5 | Credit note | Credit £100 on a charge, then try without a reason | Credit created with a reason; refused without one | |
| K6 | Receipts import | Import a CSV of receipts with one bad reference | Good rows matched to invoices; bad row reported, nothing half-loaded | |
| K7 | Finance export | Export charges to finance; export again | File has new charges only; exported charges are locked | |
| K8 | Module off | Switch the rent ledger off; reopen Rents | Section says it is switched off; menu link gone | |
| K9 | Demands on hold | Start a demand run for a period that includes DL-0031 | The preview names DL-0031 as on hold; no rent or service charge is raised for it | |
| K10 | Take a hold off | In Edit this agreement untick Rent demands on hold on DL-0031; run the same dates again | DL-0031 is charged once; no other agreement is charged twice | |
| K11 | Arrears passed on | Rents, Arrears: filter to arrears passed to debt collectors, then to those not passed on; download the list | DL-0024 is tagged with the date and reference; each filter shows the right agreements; the download has the same rows and the referral columns | |
| K12 | Billing contact on demands | Open the rent account and a statement for DL-0001; export its charges to finance | The account and statement show the billing contact; the export uses the billing address and ends with Billing contact and Billing email columns | |

## L. Service charges

| ID | Test | Steps | Expected | Result |
|---|---|---|---|---|
| L1 | Schedules | Service charges: open both schedules | Riverside (6 leases) and Northfield Business Park (8 leases); empty units borne by the council | |
| L2 | Apportionment | Open the Northfield apportionment matrix | Shares by floor area add to 100%; DL-0509 cap applied, excess borne by the council | |
| L3 | Budget and demands | Approve next year's budget; issue the budget | Quarterly on-account demands appear on each tenant's rent account | |
| L4 | Spending | Add a cost; import spending with Category and Tenants agreed columns | Costs appear against headings; budget v actual updates | |
| L5 | Improvements and BSA defects | Save a recoverable improvement without tenants' agreement; mark a cost as a Building Safety Act defect | First refused; defect forced to not recoverable | |
| L6 | Late issue | Record statements as issued after the four-month deadline, without and with a reason | Reason required; printed on the statement with an apology | |
| L7 | Year end | Reconcile last year and open a tenant statement | Balancing charge or credit per tenant; certified-by shown | |
| L8 | Deadlines (IT) | Run the weekly deadline email | Administrators get budgets and statements due in 60 days | |

# M. Maintenance

| ID | Test | Steps | Expected | Result |
|---|---|---|---|---|
| M1 | Report a repair | Report a job on DM-N05 Community Hall | Target date set from priority; responsibility suggested from the lease | |
| M2 | Emergency job | Open the pool hall ceiling job at DM-N04 | Emergency, ordered, target today | |
| M3 | Works order approval | Open the roof lantern job (DM-N09, £14,800), waiting for Sam Hughes; as Administrator approve it, then issue the order | Order held until approved | |
| M4 | Works order | Issue an order and download it | Order names the contractor, property, job, priority and target date | |
| M5 | Complete and recharge | Complete the graffiti job on DM-N01-U06 and raise the recharge | Recharge invoice on the tenant's account | |
| M6 | Tenant's repair | Open the shop front glass job at DM-N02-U06 | Responsibility Tenant, with the lease reason | |
| M7 | Planned maintenance | Planned: filter overdue and statutory | Leisure centre water temperatures and library emergency lighting overdue; jobs raised automatically when due | |
| M8 | Components | Components: filter by building and surveyor; condition D | Library flat roof (D) listed with replacement cost | |
| M9 | Contractors | Open Clearview Glazing Northfield | Insurance expiring within 30 days flagged | |

## N. Compliance and the Golden Thread

| ID | Test | Steps | Expected | Result |
|---|---|---|---|---|
| N1 | Overview | Compliance: filter to DM-N04, then to Owen Jones | Leisure centre: TM44 missing, emergency lighting expired, actions open on the FRA and EICR. Library (DM-N03): FRA expired, EICR unsatisfactory | |
| N2 | Record a certificate | Upload a new emergency lighting certificate for DM-N04 | Status turns Valid; expiry calculated from the inspection date | |
| N3 | Remedial actions | Open the DM-N04 fire risk assessment | One action open (escape route), one done; mark it done | |
| N4 | EPC below E | Report EPC rated F or G | DM-N01-U04 (G), DM-N02-U07 (F) and DM-E01-U03 (F) listed | |
| N5 | Golden Thread tier | Open DM-N03 building safety | Tier B, height, responsible person and who confirmed it | |
| N6 | Building documents | Upload a new version of the DM-N03 asbestos register | Old version kept as superseded; Before you visit links the new one | |

## O. Asset valuation

| ID | Test | Steps | Expected | Result |
|---|---|---|---|---|
| O1 | Programme | Asset valuation: this year's programme | Every classified asset due this year listed, with assignee | |
| O2 | Assign | Assign DM-N09 to the external valuer, and DM-N03 to Lena Chen | Assignee emailed; appears on their My work | |

| ID | Test | Steps | Expected | Result |
|---|---|---|---|---|
| O3 | Record and accept | Enter a fair value with land and buildings that do not add up; correct; accept as Administrator | Refused until they add up and the fair value level is set; accepted ones are locked | |
| O4 | Indexation | Open DM-N04 (valued 2022) | Indexed value to the year end shown with the factor | |
| O5 | Audit query | Answer the open audit query on the market hall, report it, then close | Statuses move to reported and closed; auditor's due date shown | |
| O6 | Year-end schedule | Download the schedule for finance | Every asset, value, fair value level and indexed value | |

# Tests P–S: data in and out, administration

## P. Reports

| ID | Test | Steps | Expected | Result |
|---|---|---|---|---|
| P1 | Ready-made reports | Run every ready-made report once | Each runs without error; module reports appear only when the module is on | |
| P2 | Check figures | Run Rent roll, Lease expiries in 12 months, Certificates expired; compare with the records | Rows and totals agree with the source records | |
| P3 | Build a report | New report on Leases: pick columns, number their order, filter rent at least £15,000 and expiry in the next 2 years, sort, show totals | Correct rows, column order and totals | |
| P4 | Blank filter | Properties report: UPRN is blank | Properties without a UPRN only | |

| ID | Test | Steps | Expected | Result |
|---|---|---|---|---|
| P5 | Download | Download P3 as Excel and CSV | Same rows as the screen; every row, not just the first page | |
| P6 | Share and copy | Share P3; as another Surveyor copy and change it | Colleague can run and copy, not edit the original | |
| P7 | Filters on list pages | On components, compliance and planned maintenance, filter by building DM-N01 and by surveyor | Only records on that building, or that surveyor's properties | |
| P8 | Lease documents report | New report on Lease documents: filter to a review date before today; download it | Lists the overdue documents from B13 with their agreement, type and review date | |
| P9 | New columns | In reports on Properties and on Leases add Portfolio, Estate strategy section, Holding purpose review date, Time is of the essence, Rent demands on hold and Estimated rental value | Each column is offered and is filled where the record has a value (for example DL-0001 and DL-0031) | |

## Q. Imports and exports

| ID | Test | Steps | Expected | Result |
|---|---|---|---|---|
| Q1 | Templates | Import: download each template | Spreadsheet with headings, guidance and an example row | |
| Q2 | Property import | Import five new properties including Site hazards; then the same file with one change | First adds five; second updates one; nothing duplicated | |
| Q3 | Validation | Import a file with a missing reference, a bad date and an unknown surveyor | Every problem listed by row; nothing imported until fixed | |

| ID | Test | Steps | Expected | Result |
|---|---|---|---|---|
| Q4 | Leases and parties | Import parties, leases and breaks in one workbook | Records linked; events created | |
| Q5 | Valuer's figures | Import returned valuation figures for two assets | Values recorded as received | |
| Q6 | Components | Import a component list for DM-N03 | Components appear with lifecycle dates | |

## R. Open data and API

| ID | Test | Steps | Expected | Result |
|---|---|---|---|---|
| R1 | Transparency Code list | Download the land and assets file as CSV, JSON and GeoJSON | Northfield and Exampleton assets with UPRN and grid reference; DM-S09 left out | |
| R2 | Available property | Download available property | Vacant lettable units only, such as DM-N01-U04 and DM-N02-U03 | |
| R3 | API | Open API docs; call the properties endpoint signed in, then signed out | Data returned when signed in; refused when not | |

## S. Change log, settings and administration

| ID | Test | Steps | Expected | Result |
|---|---|---|---|---|
| S1 | Change log | Edit a lease rent; open Changes | Who, when, before and after values shown; filter by person works | |
| S2 | Supervisors | Set a supervisor on a user record | Their approval requests go to that supervisor | |
| S3 | Branding | Change the organisation name and brand colour | Header updates for everyone | |
| S4 | Modules | Switch maintenance off and on | Menu, dashboard panels and reports follow the setting; no data lost | |

| ID | Test | Steps | Expected | Result |
|---|---|---|---|---|
| S5 | Case types | Change which stages need approval for one case type | New cases follow the new setting | |
| S6 | Help pages | On a property, a case and the dashboard, select Help with this page; search Help for "hand over" | The matching help page opens with screenshots; search finds Users and roles | |
| S7 | Training courses | Help, Training courses; open the System administrator course and complete the Users module | Steps show highlighted screenshots, including Hand over work; the exercise answer (Sam Hughes) and the quiz are marked correctly; progress is kept after reloading | |
| S8 | Two-step rule | In Manage data set two-step sign-in to required for everyone; sign in as a Viewer; set it back | The Viewer is asked to set up two-step sign-in before doing anything else | |
| S9 | Test account requests | On a public test site only: signed out, select Request a test account and send the form; as Administrator open Manage data, Test account requests and choose Approve as surveyor | The person is emailed their username and a one-time link; after choosing a password they sign in as a Surveyor | |

## Tests PJ: projects

Run these on days 4–5 with casework. The demo data has three projects: PRJ-2026-001 Northfield Business Park phase 2 (a development led by Lena Chen), PRJ-2026-002 Depot site acquisition, Mill Lane (confidential, led by Rob Baker, with Jo Patel and Sam Hughes on the team) and PRJ-2026-003 disposal of surplus land at Oak Street and Lower Farm. Reload the demo data afterwards: PJ7 and PJ12 change it.

| ID | Test | Steps | Expected | Result |
|---|---|---|---|---|
| PJ1 | Start a project | As a Surveyor, Projects, Start a project; choose Acquisition, give a proposed site and title number | Created with the next PRJ reference, in Idea, confidential, with starter tasks | |
| PJ2 | Who can start | As a Viewer, type `/projects/new/` | Refused | |
| PJ3 | Confidential | As a Surveyor not on PRJ-2026-002's team, open Projects, a DM-N01 property page and Reports, Projects; type its address | It appears nowhere; the address shows page not found | |
| PJ4 | Access log | As an Administrator, open PRJ-2026-002 | Who has opened it lists each opening, including yours | |
| PJ5 | Gateways | On your PJ1 project, record gateway 1 Approved, then gateway 2 Deferred, then Rejected | Moves to Business case; stays there; then Stopped with the reason | |
| PJ6 | Milestone status | On PRJ-2026-001, set a milestone's target to yesterday | The project shows red, Milestone missed | |
| PJ7 | Budget | On PRJ-2026-001, enter figures for a year with forecast above approved | Totals by financial year and "over budget" shown | |
| PJ8 | Who sees the budget | Give a Surveyor test account only Estates, then only Finance; open PRJ-2026-001 | Budget hidden with Estates only; shown and editable with Finance | |
| PJ9 | Spending import | As a finance officer, Import, Project spending, with a row for PRJ-2026-001 and a made-up budget line | The made-up line is rejected by name; a correct row updates the actual figure | |
| PJ10 | Risks and linked work | Add a risk with likelihood 4 and impact 5; link an open case by reference | Score 20, High, listed first; the case page shows Part of project | |

| ID | Test | Steps | Expected | Result |
|---|---|---|---|---|
| PJ11 | Proposed site | On your PJ1 project, Draw the proposed site with one known INSPIRE ID and one made-up ID | The made-up ID is named and nothing changes; with known IDs only, the boundary shows on the map | |
| PJ12 | Completion checklist | Open PRJ-2026-003; follow Record the disposal for DM-S06 and save a completion date and price | Checklist ticks that step itself; DM-S06 shows Disposed of and is gone from the open data download | |
| PJ13 | Prefilled forms | Open PRJ-2026-002's checklist and select Add the property and Add the title | The forms open with the site address and title number filled in; nothing is saved until you save | |
| PJ14 | Hand over | Hand over work for Rob Baker | Projects he leads, his team places, project tasks and risks are listed and can go to another Surveyor | |
| PJ15 | My work and phone | Add a Surveyor test account to PRJ-2026-001's team and give it a milestone, then open My work as that account; then a project page at phone width | My projects and project milestones appear; the project page fits without sideways scrolling | |

## Tests T–Z: cross-cutting checks

Run these on a sample of pages: the dashboard, a property, an agreement, a case, a job, a report and an import.

# T. Accessibility (WCAG 2.2 AA)

| ID | Test | Steps | Expected | Result |
|---|---|---|---|---|
| T1 | Keyboard only | Use Tab, Shift+Tab, Enter and Space only | Every link, button, menu and panel reachable; yellow focus always visible; Escape closes header menus | |
| T2 | Screen reader | NVDA (Windows) or VoiceOver (Mac): read a case and the dashboard | Headings in order; panel buttons say expanded or collapsed; tables read with headers | |
| T3 | Zoom and reflow | Zoom to 400% at 1280px wide | No sideways scrolling except wide tables, which scroll on their own | |
| T4 | Colour | High contrast theme; and a colour-blindness simulator on the Clear style | Nothing relies on colour alone; tags and panels carry words | |
| T5 | Forms and errors | Submit a lease form with errors | Error summary at the top links to each field | |
| T6 | Automated check | Run axe DevTools or WAVE on the sample pages, at desktop width and at phone width (390px) | No serious or critical issues, including target size | |
| T7 | Scrolling tables by keyboard | At phone width open Properties; Tab to the property table and use the arrow keys | The table box takes focus with a yellow outline and scrolls sideways; a screen reader announces it by the table's heading | |

## U. Security and permissions

| ID | Test | Steps | Expected | Result |
|---|---|---|---|---|
| U1 | Role boundaries | Repeat A3 to A6 by typing edit and action addresses directly | Every forbidden action refused (403), not just hidden | |
| U2 | Files | Copy a document link; open it in a private window | Sign-in required | |
| U3 | Uploads | Upload an .exe renamed .pdf, and a file over the upload size limit | Both refused with a clear message | |
| U4 | Confidential data | As Contributor open a case with confidential notes; check DM-S09 in open data | Notes hidden; DM-S09 never published | |
| U5 | Session | Leave a session idle beyond the timeout; sign out and press Back | Asked to sign in again | |
| U6 | Passwords | Set a short or common password | Refused | |
| U7 | Behind HTTPS (IT) | On the hosted test server (HTTPS proxy, `BEHIND_HTTPS_PROXY=True`, `SECURE_COOKIES=True`, `https://` `PHAEMA_BASE_URL`): sign in, save a property, sign out | All work; no "Forbidden (403) CSRF" message; the browser shows the session cookie as Secure | |
| U8 | Plain HTTP refused (IT) | On the same server open the `http://` address | Redirected to `https://` by the proxy; the app port 8000 cannot be reached from outside | |
| U9 | API and two-step sign-in | Call the API with the username and password of an account that uses two-step sign-in | Refused | |
| U10 | Reset link limits | Ask for a password reset for the same account six times within an hour | No more than three emails arrive; the sixth request shows a Too many requests message | |

## V. Browsers and devices

| ID | Test | Steps | Expected | Result |
|---|---|---|---|---|
| V1 | Desktop browsers | Run B, C2, G7 and K1 in Edge, Chrome, Firefox and Safari | Same behaviour in each | |
| V2 | Phone | On a phone open My work, Whole estate, Properties, a property, the map and Hand over work | The page never scrolls sideways; wide tables scroll inside their own box; menu and setting links are easy to tap | |
| V3 | Print | Print a case, a property and a tenant statement | Clean print without menus; hidden panels printed | |

## W. Performance

| ID | Test | Steps | Expected | Result |
|---|---|---|---|---|
| W1 | Page speed | Time the dashboard, property list, map and a large report | Each under 3 seconds on the hosting server | |
| W2 | Large download | Download the change log and rent schedule to Excel | Completes; every row included | |
| W3 | Several users | Five testers use the system at once for 15 minutes | No errors or noticeable slowdown | |

## X. Regression and sign-off

| ID | Test | Steps | Expected | Result |
|---|---|---|---|---|
| X1 | Retest | Rerun every test that failed, after its fix | Passes | |
| X2 | Automated suite (IT) | Run `python3 -m pytest` on the final build | All tests pass | |
| X3 | Fresh install (IT) | Install from the installation guide (week 8) on a clean server; load the demo data | Works first time following the guide only | |

## Y. Hand over work

Run these last: they move the demonstration surveyors' work, which earlier tests rely on. Sign in as an Administrator. Lena Chen, Rob Baker and Sam Hughes are demonstration users; you need no password for them. When finished, reset the demo data (Y9).

| ID | Test | Steps | Expected | Result |
|---|---|---|---|---|
| Y1 | Who can use it | As Surveyor and as Viewer, look for Hand over work in the menu, then type `/people/hand-over/` | Not in their menu; the address is refused (403) | |
| Y2 | What a person holds | Hand over work; from Lena Chen; Show their work | 19 properties, 19 open cases, 96 case tasks, 24 lease events, 3 maintenance jobs; a note that she is also an internal valuer | |
| Y3 | Part of a patch | From Lena Chen, give it to Rob Baker, only work on DM-N01-B01 Hangar block; Show their work, then Hand over work | Counts first show 6 properties, 5 cases, 29 tasks, 10 lease events, 1 job; staff row says not moved; after, those move to Rob and Lena keeps 13 properties | |
| Y4 | Different people per kind | From Lena Chen: properties to Rob Baker, open cases to Sam Hughes, tasks left with Lena | A moved case shows Sam as owner; its tasks still show Lena; her properties show Rob as managing surveyor | |
| Y5 | Approvals and staff | From Sam Hughes, everything to your own account | 2 case approvals, 2 works orders and 5 staff move; you are emailed about each case approval (or it appears in the log); the five surveyors' user records show you as supervisor | |

| ID | Test | Steps | Expected | Result |
|---|---|---|---|---|
| Y6 | A leaver | From Lena Chen, give it to Rob Baker, tick Also stop signing in; Hand over work | Success message lists what moved; every count for Lena is now 0; her user record is no longer Active | |
| Y7 | Change log | Changes, filtered by your name | Each moved property, case, task, event and job is listed, with the old and new person | |
| Y8 | Who can receive | From any surveyor, open each Give to list | Viewers never appear; the Contributor (Jo Patel) appears only for open cases they contribute to | |
| Y9 | Reset (IT) | `./phaema.sh demo-data --reset` | Lena Chen's counts are back to those in Y2 | |

## Z. Access areas and cover

Run these on day 11 with the other cross-cutting tests. Use the demonstration staff: Lena Chen and Rob Baker (Estates, Maintenance and compliance, Valuation), Sam Hughes (their supervisor), Dana Okafor (finance officer: Surveyor role, Finance area) and Jo Patel (Legal). Sign in as an Administrator to change areas; the others have no password, so test what they see by giving your own test accounts the same areas.

| ID | Test | Steps | Expected | Result |
|---|---|---|---|---|
| Z1 | Standard areas | Manage data, Access areas | Estates, Finance, Maintenance and compliance, Valuation and Legal, with the modules shown | |
| Z2 | Module refused | Give a Surveyor test account only Legal; open the menu, then type `/maintenance/jobs/` and `/rents/` | Buildings and Income menus gone; both addresses show "You do not have access" | |
| Z3 | Module opened | Add that account to Maintenance and compliance | Maintenance and compliance open; rents still refused | |

| ID | Test | Steps | Expected | Result |
|---|---|---|---|---|
| Z4 | Own patch | Give the account Estates and make it managing surveyor of DM-N01; open Rents, Arrears and a DM-N01 lease, then a lease elsewhere | Rents says "Showing rent for your patch"; arrears list only DM-N01 leases; the other lease has no rent panel and its account address is refused | |
| Z5 | Whole chain | Make your account the supervisor of Sam Hughes; open Rents | Lena Chen's and Rob Baker's tenants now appear | |
| Z6 | Cover | As Sam's supervisor, select Arrange cover for absence on My work and look at Who is away. Then, as Administrator, arrange cover for Lena Chen by another test account from today until tomorrow, and later select End now | Only people in your chain are offered to supervisors; the covering account sees Lena's arrears; End now removes it | |
| Z7 | Finance officer | Give a Surveyor account only Finance; open Rents, Raise rent, Finance exports, Import | All open; Import shows only receipts and service charge spending; Compliance refused | |
| Z8 | Finance viewer | Give a Viewer account Finance | Can see arrears and accounts; Raise rent and Import refused | |
| Z9 | Reports and change log | As the Z2 account, open Reports and Changes; try `/reports/new/?dataset=charges` | No rent, maintenance or compliance reports; no changes to charges or jobs listed; the address is refused | |
| Z10 | Approvals and hand-over | Remove Maintenance from a supervisor, then ask them to approve a large works order; open Hand over work for a surveyor | The order is refused with a message naming the supervisor; maintenance jobs can only go to people with Maintenance | |

| ID | Test | Steps | Expected | Result |
|---|---|---|---|---|
| Z11 | Change log | Changes, filtered by your name | Area changes and cover appear with who, when and what | |

## Defect log and sign-off

Log each failure here or as a GitHub issue labelled `testing`, newest first.

| No. | Test | Severity | What happened | Raised by, date | Fixed in | Retested |
|---|---|---|---|---|---|---|
| 1 | | | | | | |
| 2 | | | | | | |
| 3 | | | | | | |

| Sign-off | Name | Date | Decision |
|---|---|---|---|
| Test lead | | | |
| Head of Estates | | | |
| IT / hosting | | | |